Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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Article history

Received:
xx xxx 2023

Received in revised form:
xx xxx 2023

Accepted:
xx xxx 2023

Published online:
xx xxx 2023

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Abstract

Malaysian social media discourse mixes English and Malay within single utterances, producing the contact varieties known as Bahasa Rojak and Manglish. No existing sentiment corpus for this variety is simultaneously large-scale, polarity-balanced, casually authentic and human-validated, and no settled practice exists for demonstrating that any such corpus is reliably labelled. This review argues that the binding constraint on progress is not model capacity but the annotation label itself, together with the credibility of the evidence that it is sound. Its scope is narrow: annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data, excluding comparative benchmarking of classifiers. Three contributions follow. A taxonomy of annotation strategies shows that each available tier secures either fidelity or scalability by conceding the other. A diagnostic account then separates neutral-label skew into a structural cause, the default return of a scoring function that recognises no sentiment-bearing token, and genuine annotation ambiguity, and pairs this with an audit of what four regional corpora report about their own reliability. No corpus reports the full set of elements a reliability claim requires, the highest reported coefficient measures a narrower task than the resource is used for, and one manually annotated corpus reports no agreement statistic at all. The third contribution establishes that the annotation defect and the measurement instrument are coupled: the class skew that corrupts the labels is the same condition under which the standard agreement statistic ceases to be interpretable, so the defect disables the instrument that would have detected it. This is why the problem is unresolved rather than merely difficult. The review closes with a research agenda comprising minimum reporting standards, skew-robust agreement measurement, and the untested premise that annotators of different construction fail independently.

Keywords: *sentiment analysis, code-mixed text, Bahasa Rojak, annotation reliability, inter-annotator agreement*

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1. Introduction

Malaysian digital discourse is not written in discrete monolingual streams. On platforms such as X, Facebook and WhatsApp, bilingual users alternate between English and Malay within a single utterance to manage efficiency, signal identity and negotiate social distance, producing the contact varieties known as **Bahasa Rojak** and **Manglish** [1], [2]. This point is worth establishing before any computational claim is made, because these varieties are routinely mischaracterised. They are not degraded English. They are **rule-governed contact varieties** with their own syntactic and pragmatic conventions, in which switching follows describable structural regularities rather than occurring at random [1], [2]. The difficulty such text presents to natural language processing is therefore not that it is careless writing. It is that the writing is systematic in ways current tooling was not built to read, and, as this review argues, in ways current *labelling* practice is not built to record.

When sentiment systems underperform on this material, the diagnosis offered is almost always **architectural**. The model, it is said, cannot represent code-switched input; a larger, better pre-trained or more regionally adapted model should therefore represent it better. That reasoning is intuitive, and it is not so much wrong as premature. This review argues that the **binding constraint** on Malaysian Malay-English code-mixed sentiment analysis lies upstream of the model. It is the **annotation label**, the quality of the supervision signal a model is trained and evaluated on, together with the credibility of the evidence offered that the signal is sound.

Two observations motivate that reframing, and both recur throughout the sections that follow. The first concerns the **corpora themselves**. No existing Malay-English sentiment corpus is simultaneously large-scale, polarity-balanced, casually authentic and human-validated; each available resource secures some of these properties only by conceding others (Section 5.2). The largest general-domain resource, MESocSentiment, illustrates the cost of that concession most starkly: its 19,714 posts are distributed **78.23% neutral, 16.33% positive and 5.44% negative** [3]. A skew of that magnitude is not a property of Malaysian discourse. It is an **artefact of the labelling method**, and Section 3.2 traces the mechanism that produces it.

The second observation concerns what the literature does not report, and it is the half of the problem that has received least attention. For the automated and semi-automated corpora, precisely those whose labels are most in doubt, **reliability is largely unreported**, and where an agreement statistic is given, it is measured under conditions that make the resulting number difficult to interpret (Section 5). The field therefore lacks not only trustworthy labels but a settled practice for demonstrating that any given set of labels is trustworthy. **Annotation and evaluation fail together**, and it is that conjunction, rather than either deficiency alone, which explains why the problem has proved so persistent.

Scope. This review examines one question: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data remains unresolved. Linguistic description, model architecture and application domain

appear only where they bear on that question, and are kept deliberately brief where they do not. Comparative benchmarking of classifiers is out of scope, as is sentiment analysis as a general field; both are well surveyed elsewhere.

It is equally important to state what this review does not do. **It does not propose, implement or evaluate a system, and it does not report the author's own work.** Its contribution is **diagnosis and direction**: an account of why the problem has resisted solution, and of what the field would need to establish before a solution could credibly be claimed. Where the analysis identifies a vacancy in the literature, it names the vacancy and stops there.

Three contributions follow from that analysis.

1. A **taxonomy of annotation strategies** for low-resource code-mixed text, arranged along a **fidelity-scalability axis**, which shows that each tier resolves one constraint by conceding another and that no reviewed tier secures both (Section 3.1, Table 1).
2. A **diagnostic account of neutral-label skew** that separates its **structural** cause, the near-zero default of lexicon and rule-based scoring under out-of-vocabulary (OOV) input, from its **ambiguity-driven** cause, together with an **audit of what regional corpora actually report about their own reliability** (Section 3.2; Section 5.2, Table 3).
3. An account of how the linguistic, annotation and measurement challenges **compound rather than merely coexist**, culminating in the finding that the class skew which corrupts the labels is the same condition under which the standard agreement statistic ceases to be interpretable, and the **research agenda** that follows from it (Section 6.2, Figure 1; Section 6.3).

The remainder of the paper proceeds from the linguistic source of the difficulty (Section 2), through annotation practice and its failure modes (Section 3), the models employed as annotators (Section 4) and the evaluation of annotation reliability (Section 5), to a synthesis that treats these challenges as a single interacting system and sets out what the field must establish next (Sections 6 and 7).

1.1. Review Methodology

The literature was identified through structured searches of ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language variety (code-mixed, code-switching, Manglish, Bahasa Rojak, Malay-English), the task (sentiment analysis, polarity, opinion mining) and the object of interest (corpus, annotation, inter-annotator agreement, label quality). The window was 2019 to 2026, extended backwards by citation chaining from the retrieved corpus papers. The date filter was waived for inter-rater reliability methodology, since the apparatus Section 5 examines predates the window and cannot be cited from within it.

Four criteria governed inclusion. A study had to address Malaysian or closely comparable Malay-English code-mixed text, or supply annotation or reliability methodology applicable to it; report enough procedural detail for its annotation or

evaluation practice to be assessed; be peer-reviewed or a preprint with a traceable venue; and contain an annotation or reliability component. The fourth criterion is what enforces the narrow scope. Studies that only benchmark classifiers against an existing corpus were excluded, however relevant their results are to Malaysian sentiment analysis in general, because they take label quality as given and this review is concerned with whether it can be.

2. Why Malaysian Code-Mixed Text Resists Reliable Annotation

Three properties of Malaysian code-mixed writing bear directly on the annotation task. All three are well described in the sociolinguistic literature; what matters here is not the description but the consequence each carries for a person or a model asked to assign a sentiment label. Each is therefore traced to that consequence and no further.

Switching is intrasentential, and the switch point carries no structural cue. Malay-English switching is dominated by patterns that operate inside the clause rather than between clauses. In Treffers-Daller et al.'s (2022) analysis of 1,044 switched utterances, **insertion** - the embedding of a content word or phrase from one language into a stretch of the other, accounts for 33.0% of switches and **congruent lexicalization** - where the grammar and lexis of both languages interact within a shared word order, for a further 30.7%, against 24.8% for clause-level **alternation**. The explanation the authors give is typological rather than stylistic: both languages share **Subject-Verb-Object (SVO)** surface order, and Malay carries very little inflectional morphology, so a speaker can interleave vocabulary mid-clause with almost no grammatical friction. Their corpus is spoken data from two teachers, so its proportions should not be transferred wholesale to social media text; the structural explanation, however, is a property of the language pair rather than of the modality, and it holds wherever the two languages meet. A representative written case is the tweet *"Does institut kanser offer free medication cheap chemotherapy"*, where a Malay noun phrase in Malay head-initial order sits inside an English interrogative frame [3].

Annotation consequence. Because the switch is not aligned to any clause boundary, the annotator cannot use a **language boundary to locate the sentiment-bearing span**. In monolingual annotation the unit carrying affect is usually identifiable before it is judged; here, identifying it is itself an interpretive act. **Span selection therefore becomes a source of disagreement prior to, and independent of, disagreement about polarity**, two annotators may assign different labels not because they read the sentiment differently but because they are reading different parts of the utterance.

Affect is carried by function words that no standard lexicon contains. Across most code-switching corpora, closed-class items resist switching. Malaysian usage inverts this: Treffers-Daller et al. [1] describe the high frequency of switched function words as one of the most remarkable characteristics of the language pair, and in their data personal pronouns account for 10.4% of switches and discourse markers for a further 9.9%. Sentence-final particles attach to otherwise-English clauses, the form "Buy this lah" is typical, and they soften an assertion or build rapport, carrying pragmatic and affective weight rather than propositional content

[1], [2]. Both sources describe speech, transcribed conversation in one case and scripted television dialogue in the other, so the pragmatic function of these particles is better evidenced than their distribution in writing. These particles appear in neither standard English nor standard Malay sentiment lexicons.

Annotation consequence. The reading of such a particle is **underdetermined by the text**. One annotator takes *lah* as a softener and downgrades the intensity of the surrounding clause; another takes it as a marker of exasperation or sarcasm and inverts the polarity. Both readings are defensible, which is precisely the difficulty. The resulting disagreement is **systematic rather than random**: it recurs on the same construction, in the same direction, for the same annotator. This distinction matters more than it may appear, and it is the reason the standard remedy fails. **Random disagreement is averaged away by recruiting more annotators; systematic disagreement is reproduced by them.** Adding annotators to a corpus of particle-heavy text buys precision in the wrong place, and it depresses the agreement statistic without indicating any deficiency in the annotators themselves.

Phonetic orthography multiplies the surface forms of a single word. Users spell as they speak rather than to a dictionary standard, so one concept surfaces across many written variants, the Malay negator *tidak* appears as *tak*, *tk* or *x*. Maskat and Rahman [4] resolve this apparent disorder into five recurring categories of social-media Malay, but report normalising only two of them by similarity matching, leaving the remainder as open research. Rule and dictionary-based normalisation additionally struggle to keep pace with slang that changes faster than the resources describing it.

Annotation consequence. The effect is not that annotation becomes harder in general, but that **human and machine annotators fail on different inputs**. Where a human reader recognises *x* as *tidak* without effort and proceeds to judge the sentiment, a lexicon or tokeniser encounters an **out-of-vocabulary (OOV)** token and returns no signal at all. The two annotator types are therefore not noisy versions of one another; their errors are **differently distributed over the data**. This asymmetry is recorded here as an observation about failure modes rather than developed further, it becomes one link in the compounding account of Section 6.2.

Taken together, these three properties attack **three different stages of the annotation process**: the first destabilises the identification of what is to be judged, the second destabilises the judgement itself and does so in a way more annotators cannot repair, and the third makes the reliability of a label depend on which kind of annotator produced it. This is why Malaysian code-mixed text is not merely difficult to process but **difficult to label consistently**, and why the difficulty propagates into every stage that depends on those labels.

3. Annotation Practice and Its Failure Modes

3.1. Annotation Strategies and Trade-offs

For low-resource code-mixed text, **fidelity and scalability pull in opposite directions**. A corpus can be labelled carefully or it can be labelled at volume, and the strategies reported in the literature are best understood not as competing

solutions to one problem but as different choices about which of the two to concede. Arranged that way, they fall into three tiers.

Tier 1: manual annotation. Labelling by fluent bilingual annotators remains the fidelity standard, and the regional corpora built this way are the only ones whose reliability is demonstrated rather than assumed. Kasmuri and Basiron [5] report a Cohen's kappa of 0.83 over 4,191 annotated sentences, of which 2,316 entered the corpus, and Kong et al. [6] hand-annotated 11,568 tweets drawn from a pool of 108,246 collected during the pandemic. What both illustrate is the cost of the choice rather than a deficiency in the method: the labour that produces high agreement is precisely what prevents these corpora from reaching the scale, or the topical breadth, that a general-purpose Manglish resource requires. Manual annotation does not fail. It simply stops.

Tier 2: semi-automated annotation. The second-tier trades in the opposite direction, using models to accelerate labelling and accepting **model bias** as the price. Mohd Shamsuddin et al. [3] exemplify the approach in constructing MESocSentiment: automated language-identification libraries filtered code-mixed posts from a pool of 229,566 tweets down to 20,226, after which an ensemble of three sentiment analysers assigned labels by agreement. The authors are explicit that the motivation is economic, automated language identification "reduced the need to find fluent speakers in Malay and English," and the sentiment analysers "eliminated the need to find speakers fluent in Malay and English to label the dataset." The reasoning is defensible on its own terms, and there is evidence from other labelling tasks that automated annotators can rival human crowd-workers [7]. The concession is that the resulting labels are **only as competent as the models producing them**, and inherit whatever those models cannot see, a cost Section 3.2 quantifies.

Tier 3: fully automated annotation with generative models. The third-tier replaces shallow classifiers with generative LLMs used directly as annotators, reasoning over context rather than matching surface forms, and requiring no task-specific training data. In principle this **dissolves the trade-off**: contextual competence at machine scale. Two findings complicate that reading, and the first comes from within the tier itself. Negi et al. [8], attempting to build fully autonomous LLM annotators for fine-grained opinion analysis, report a **performance bifurcation**, their models were reliable at identifying opinion spans but struggled to reproduce the relational structure connecting those spans, and conclude that LLMs are better positioned as "high-fidelity annotation assistants and data augmentation tools… rather than replacing human annotators entirely." The exemplar of full automation, in other words, declines the claim of full automation. The second finding concerns transfer: evaluating LLM annotators on Marathi, Jadhav et al. [9] found GPT-4o and Llama 3.1 405B trailing a fine-tuned BERT baseline by **10.2% and 14.1% accuracy** respectively, and concluded that fine-tuned transformers remain necessary where linguistic resources are thin. Neither study examines Malaysian code-mixed text, and Marathi is a morphologically rich language where Malay is not, so these results bound the tier's promise rather than measuring its performance in this setting. That they bound it at all is the point: the evidence that generative annotation dissolves the trade-off does not yet exist, and the two most direct attempts to produce it report otherwise.

Table 1. Annotation Strategies for Low-Resource Code-Mixed Text

Tier	Representative Work	Fidelity	Scalability	Contextual Robustness	Domain Failure Mode
Manual	Kasmuri & Basiron [5]; Kong et al. [6]	High, but demonstrated only for binary subjectivity (κ = 0.83), not polarity	Low	High	Coverage; corpora stay small and domain-bound
Semi-automated (model voting)	Mohd Shamsuddin et al. [3]	Not established	High	Low	Shared blind spots across voters; non-majority cases removed
Fully automated (generative)	Negi et al. [8]	Uneven, strong on spans, weak on structure	High	High on well-resourced varieties	Relational error, output variance, and a low-resource accuracy penalty

Read across the rows, the pattern is that **each tier resolves one constraint by conceding another**, and that the concession is structural rather than incidental, it follows from what the strategy is, not from how carefully it was executed. Manual annotation concedes scale, semi-automation concedes the demonstrability of fidelity, and generative annotation concedes reliability exactly where linguistic resources are scarcest, which is the condition under which it would be most useful. **No strategy in the reviewed literature secures fidelity and scale together for Malaysian code-mixed text.** Whether any arrangement could be a question this review does not attempt to settle; it returns in Section 6.3 as a direction for the field, and is left open there.

3.2. Neutral-Label Skew

The clearest regional evidence of what semi-automated labelling costs is the class distribution of the corpus it produces. MESocSentiment, the largest general-domain Malay-English resource, is distributed **78.23% neutral, 16.33% positive and 5.44% negative** across 19,714 posts [3]. It is worth stating at once what this figure is and is not. It is **not** a finding about Malaysian discourse; the corpus was collected under the "#Malaysia" keyword across sports, politics and the arts, topics on which social media users are not characteristically neutral. It is an **artefact of the labelling method**, and the mechanism that produces it is visible in the corpus authors' own reporting. Two distinct causes are at work, and conflating them obscures the problem.

Mechanism 1: a structural default. Lexicon- and rule-based sentiment tools derive a **continuous polarity score** by aggregating the sentiment values of the tokens they recognise, then partition that score into discrete classes by threshold. VADER — one of the three annotators used to construct MESocSentiment — states the rule explicitly: a text is classified positive at a compound score of 0.05 or above, negative at −0.05 or below, and **neutral in the band between them** [10]. The consequence deserves stating plainly, because it is the origin of the entire problem

this section describes. "Neutral" is not a positive judgement that a text is affectively balanced. It is the **default return of a scoring function that found nothing to score**: a post containing no recognised sentiment token scores exactly zero, and zero falls inside the neutral band by construction rather than by interpretation. The class is therefore critically dependent on lexicon coverage, a dependency Selvaraju et al. [11] identify directly, observing that lexicon approaches rely heavily on the availability and quality of the lexicon and break down where it lacks entries for the words actually present. On Malaysian code-mixed input, where affect is routinely carried by Malay tokens, phonetic spellings and particles absent from an English lexicon (Section 2), those tokens contribute no signal, the aggregate is pulled toward zero, and a strongly opinionated post is returned as neutral, a false neutral.

That this mechanism actually fires can be shown from the MESocSentiment paper itself. Alongside their corpus the authors present a small set of tweets whose automatic labels they checked by hand, and **all three of the errors they exhibit are false neutrals**, each a positive post labelled neutral. What makes the sample instructive is not the direction of the errors alone but where the sentiment sits in each case. The three misclassified posts carry their polarity in Malay: *memang cantik sangat* ("really very beautiful"), *selamat deepavali* ("happy Deepavali"), *raih gelar juara* ("won the champion title"). The correctly classified negative example in the same paper carries its polarity in English, the word *worthless*. Six author-selected examples are illustrative rather than statistical, and no error rate can be computed from them. But their uniformity is difficult to dismiss: where the ensemble succeeded, the affect was in the lexicon's language; in every case where it failed, the affect was not.

Mechanism 2: genuine annotation ambiguity. The second cause is unrelated to tooling. Across code-mixed corpora, neutral and mixed-feelings are consistently the hardest categories for human annotators and classifiers alike, because the boundary between an objective statement and a weakly polarised one is intrinsically blurred [12]. Some posts are neutral, and some are genuinely hard to call.

The two mechanisms produce identical labels and require opposite responses, which is why separating them matters. A **truly neutral** post is correctly labelled and belongs in the corpus. A **falsely neutral** post is a labelling error that a better-resourced annotator would not make. A single-model pipeline is structurally unable to tell them apart, because the evidence distinguishing them, that the post contains affect the annotator could not read, is precisely the evidence that annotator lacks.

The consequence of failing to separate them is more damaging than the headline percentage suggests. **A false neutral does not merely inflate one class; it depletes another.** Every strongly polarised post absorbed into the neutral class is a training example removed from the minority polarity that a downstream classifier most needs, so the same error simultaneously enlarges the majority class and erodes the minority ones. At a 5.44% negative share, the class least represented in the corpus is also the one most exposed to this erosion.

Why additional voters do not repair it. The intuitive remedy is consensus, and MESocSentiment is evidence against the intuition, because it already was a consensus pipeline. Three sentiment analysers of nominally different construction,

a rule-based system, a lexicon-based system and a pretrained transformer, voted on every post, and the corpus skewed anyway. The reason is that **voting cancels error only to the extent that voters fail independently**. Where all three resolve sentiment through surface lexical matching, none of them fine-tuned on Malaysian material, they do not merely make similar errors; they make the same error on the same inputs. A Malay affective token invisible to one is invisible to all, so the ensemble converges on a false neutral with unanimity, and unanimity is indistinguishable from confidence.

This exposes a premise on which the consensus literature rests without examining it. The claim that **architecturally different annotators fail independently** is what licenses aggregation in the first place, and on code-mixed input it is asserted rather than measured; no reviewed work reports the correlation of errors between annotator types on this material. The observation above is suggestive that independence fails for shallow lexical voters, but suggestion is not measurement, and the review does not treat it as such. It is recorded here as an **unverified premise** and returned to in Section 6.3, where establishing whether it holds is set out as a question for the field rather than answered.

3.3. Consensus and Disagreement

Sections 3.1 and 3.2 established that single-model labelling is unreliable on this material and that pooling several models does not automatically repair it. This section examines consensus as a method in its own right, organised along the axis that actually separates the available frameworks: **how each treats disagreement between voters**. Read that way, the literature traces a progression from discarding disagreement, to stabilising it, to reasoning through it.

Discard: static frequency voting. The baseline is the pipeline used to construct MESocSentiment [3]. Three sentiment analysers label each post independently, and their outputs are compared: where at least two agree, that label is assigned; where all three differ, the post is withdrawn from the corpus. The composition of the ensemble is set out in Table 2.

Table 2. Architectural Composition of the Baseline Voting Ensemble [3]

Voter	Architecture	Sentiment basis	Contextual and regional capacity
VADER	Rule-based	Heuristic polarity rules tuned for general social media	None; no code-mixed awareness
TextBlob	Lexicon-based	Dictionary pattern-matching against a sentiment lexicon	None; English-centric lexicon
pysentimiento	Transformer-based	Pre-trained deep model for social NLP	Limited; not fine-tuned on Malay or code-mixed text

The design intent is sound. Combining a rule-based, a lexicon-based and a transformer-based system is a recognised way of ensuring that voters do not share a single systematic error. Two weaknesses undermine it in practice.

The first is that the diversity is nominal rather than functional. As Table 2 shows, all three voters resolve sentiment from surface features, and none has been

fine-tuned on Malaysian material. They differ in implementation while agreeing in competence, which is the property that matters for voting. Section 3.2 traced the consequence: a Malay affective token invisible to one is invisible to all, so the ensemble does not cancel the error, it ratifies it.

The second weakness is sharper, and concerns what happens to the posts the ensemble cannot agree on. Where all three analysers return different labels there is no majority to assign, and the authors state their response plainly: "ambiguous cases were removed to ensure the corpus only contains tweets from absolute categories." This affected **512 posts, 2.53% of the 20,226 labelled**. The difficulty is not the proportion but the composition. A post on which a rule-based, a lexicon-based and a transformer-based system all disagree is, almost by definition, one where the sentiment signal is contested, ambiguous or carried by material none of the three reads confidently. These are the heavily code-switched, particle-laden, phonetically spelled cases identified in Section 2 as the hard core of the problem. **The selection rule therefore removes precisely the examples a robust classifier would most need to learn from**, and it does so systematically rather than at random.

The effect on the corpus is measurable, and it runs in the wrong direction. Before the contested posts were withdrawn, neutral accounted for 76.23% of the 20,226 labelled posts; afterwards it accounted for 78.23% of the 19,714 that remained. The withdrawal therefore left the corpus roughly two percentage points **more** skewed than the voting stage had produced (a figure computed from the paper's own tables rather than reported in it). Taken together with the first weakness, static voting has an unfortunate double effect: it **retains the labels it is confident about and wrong about, and deletes the ones it is uncertain about and would have learned from**.

Stabilise: multi-attempt sampling. A first refinement changes what is being aggregated. Rather than polling several models once, Niimi [13] polls a single locally hosted, quantised LLM repeatedly and takes the majority across attempts, motivated by the observation that the variability and reproducibility of LLM output across trials are rarely considered in annotation work. The finding is a useful one about how to spend a fixed compute budget: majority voting over multiple attempts with a medium-sized model produced more robust results than a single attempt with a large one. The gain is real, but it addresses a different failure from the one at issue here. Repeated sampling **reduces the variance of one annotator; it does not introduce a second kind of annotator**, so a blind spot shared across attempts is reinforced rather than exposed. Nor does the approach engage the tie problem, since disagreement between architectures never arises within it. The evaluation is also on English restaurant reviews, so its transfer to code-mixed input is untested.

Adjudicate: disagreement as information. The most capable approach treats conflict as evidence to be interpreted rather than noise to be averaged or discarded. Negi et al. [8] pair a declarative annotation pipeline, which reduces the variability introduced by ad hoc prompt engineering, with a dedicated procedure in which an LLM adjudicates between competing labels and produces a final annotation. Applied to this setting, such a stage would ask **why** annotators conflict, for instance whether an apparent contradiction reflects sarcasm or emphatic Bahasa Rojak, and resolve the case rather than purge it. The principle is corroborated outside the

region: in a controlled comparison, a disagreement-centred adjudication stage allowed non-reasoning models to match or exceed reasoning models relying on self-verification alone [14].

The costs should be stated with the same candour, though the direction of the comparison matters. Adjudication adds inference passes and is therefore more expensive per label than single-pass voting, but Ahtisham et al. [14] report that orchestration of this kind reaches comparable reliability at roughly a fifth fewer tokens than relying on reasoning-capable models alone, so whether adjudication is costly depends on what it is being compared against. An adjudicator is also an annotator, and a single-LLM adjudicator inherits the low-resource weakness documented in Section 3.1 [9]. Neither the cost profile nor the reliability of adjudication has been established at corpus scale on Malaysian code-mixed data.

Along the disagreement axis, then, the progression runs **discard, stabilise, adjudicate**, and each stage addresses a failure the previous one left open while inheriting those it does not address. Static voting has no route through the contested case. Multi-attempt sampling stabilises a single annotator's output without diversifying the annotator. Adjudication reasons through conflict but carries an untested cost profile in exactly the low-resource conditions where it would be applied. **No reviewed work evaluates these mechanisms in combination on this material.** Whether combining them would help is therefore an open question rather than a demonstrated result. This review reports the vacancy and does not attempt to fill it.

4. Models as Annotation Instruments

The models discussed so far appear in this review in one capacity only: as **producers of labels**. The question is therefore not which architecture performs best, which is a question about classification and outside the scope set in Section 1, but what each kind of model **fails at when asked to annotate**, since that failure is inherited by every corpus it builds. Two families are relevant.

Fine-tuned multilingual encoders acquire their competence from exposure. Manukonda and Kodali [15] illustrate the mechanism directly: fine-tuning XLM-RoBERTa with Masked Language Modeling over original, fully transliterated and partially transliterated versions of the same data produced state-of-the-art results on mixed-script, code-mixed input, because the model was shown the variant forms during training and learned to relate them to their roots. Their task is fake news detection in Malayalam rather than sentiment in Malay, so the transfer is structural rather than direct; what transfers is the principle that **phonetic and orthographic variation is tractable once it is represented in the fine-tuning data**. The corollary follows from the method rather than from any finding the authors report: an encoder resolves the variation it has been trained on, and the coverage of a fine-tuned annotator is bounded by the coverage of the data used to fine-tune it. Extending that coverage requires more annotated or transliterated regional data, which is the resource whose scarcity motivated automated annotation in the first place.

Generative models locate their competence differently. Because they are not bound to a task-specific vocabulary, they can in principle interpret an unfamiliar

spelling or an unseen slang term from context, without that form having appeared in any training set assembled for the task. This is the capability the encoder lacks, and it is the reason generative annotation is attractive for a variety that changes faster than its resources. The competence is nonetheless unevenly distributed across languages. Jadhav et al. [9] found frontier models trailing a fine-tuned BERT baseline by 10.2% and 14.1% accuracy when annotating a low-resource language, and the underlying cause is visible in how multilingual models are built: capacity is finite and spread across languages, so per-language quality is diluted [16], and robustness to code-mixing tracks how well a language is represented in the pre-training corpus. Examining Indonesian code-mixed with four embedded languages, Adilazuarda et al. [17] found mixing with English handled markedly better than mixing with the local languages, **Malay among them**, and attributed the gap to pre-training corpus bias. Generative reasoning, in other words, is strongest where it is least needed.

The operative point for this review is not the relative standing of the two families but the **shape** of their respective failures. The encoder is reliable on what it has seen and blind to what it has not; the generative model reasons about the unfamiliar but is weakest precisely where linguistic resources are thin. **Selecting an annotator is therefore not selecting a level of quality but selecting a failure mode**, and the labels a corpus contains carry the signature of whichever mode was selected. Section 6.2 draws on this framing rather than on the architectural detail.

5. Evaluating Annotation Reliability

5.1. Agreement Metrics and the Prevalence Paradox

An annotation is trusted on the strength of evidence that independent annotators, given the same text, produce the same label. The standard apparatus for that evidence is **inter-rater reliability**, expressed as a chance-corrected agreement coefficient: **Cohen's kappa** for two raters [18], **Fleiss' kappa** for three or more [19], conventionally read against the interpretive bands proposed by Landis and Koch [20]. The correction for chance is what makes these coefficients preferable to raw percent agreement. Two annotators labelling at random will agree some of the time, and the more unevenly the classes are distributed, the more often that happens; kappa subtracts the agreement expected under independence and reports only the excess.

That correction, however, is also where the instrument becomes unreliable on the corpora this review is concerned with. Kappa takes the form $(P_o - P_e) / (1 - P_e)$, where P_o is observed agreement and P_e is agreement expected by chance. When one class dominates, P_e rises toward P_o, the denominator shrinks, and the coefficient collapses even though the annotators agreed on almost every item. Feinstein and Cicchetti [21] named this the **prevalence paradox**: high observed agreement and low kappa can occur together, and when they do, the coefficient is reporting the **class distribution as much as the annotators' competence**. Prevalence-robust alternatives exist, notably Gwet's **AC1** [22], which estimates chance agreement in a way less sensitive to skew and was designed for precisely this condition.

The connection to Section 3.2 is direct, and it is the point at which the two halves of this review meet. **A corpus that is 78.23% neutral is exactly the condition under which kappa stops being interpretable.** Consider two annotators working on such a corpus who both default to neutral whenever the sentiment signal is unclear, which Section 2 established is frequently. They will agree on the overwhelming majority of items, so raw percent agreement will be high and will suggest a reliable annotation. Chance agreement on a 78% majority class is also high, so kappa will be low and will suggest an unreliable one. **Neither number describes what the annotators actually did.** The first is inflated by the skew and the second is deflated by it, and the underlying behaviour, two annotators falling back on a default, is invisible in both.

The consequence should be stated as a **reporting standard rather than a caveat**. On a heavily skewed corpus, no single coefficient carries enough information to support a reliability claim. What is required is either kappa reported together with raw percent agreement and the class prevalence, so that a reader can see which of the two effects is operating, or a prevalence-robust coefficient reported in their place. Reporting kappa alone on skewed data is not a conservative choice; it is an uninformative one.

A second limitation receives even less attention, and it concerns not which coefficient is used but **what it was computed on**. An agreement figure derived from a randomly drawn sample of a corpus describes that corpus in proportion to its composition, which means it is dominated by whatever is most common in it. For a Malaysian code-mixed corpus, the most common items are the ones with the least code-switching and the least orthographic irregularity, since heavy switching is a property of a minority of posts. **An unstratified agreement estimate therefore describes the easy majority of the corpus rather than the cases whose annotation is actually in question.** A figure of 0.80 computed this way is compatible with near-total annotator disagreement on the heavily code-switched subset, and nothing in the reported number would reveal it. A reliability estimate without a stated sampling scheme is close to uninterpretable, and, as Section 5.2 shows, stating the sampling scheme is not current practice.

5.2. What Existing Corpora Establish About Their Own Reliability

The preceding argument concerns what a reliability claim would need in order to be credible. This section asks a narrower and more answerable question: what do the regional corpora actually report about themselves? Table 3 audits the four principal Malay-English sentiment resources against the reporting elements Section 5.1 identifies as necessary. The purpose of the audit is to make the evaluation gap **visible in the literature rather than asserted about it**.

Table 3. Reporting Practice in Malay-English Code-Mixed Sentiment Corpora

Corpus	Annotators and stated qualification	Annotation guideline available	Agreement statistic reported	Sampling scheme for the estimate	Class distribution reported
MY-EN-CS [5]	Two undergraduates, screened for written and spoken Malay-English proficiency, then trained	Yes; the guideline is the paper's primary contribution	Cohen's κ = 0.83, **for a binary subjectivity judgement, not polarity**	Whole annotated set (4,191 sentences)	Subjectivity only; no polarity distribution
MyCovid-Senti [6]	Three independent annotators; **no qualification or proficiency stated**	Yes; four-question, semantic-role-based scheme	None	Not applicable	**Yes**: 48.9% negative, 27.5% positive, 23.6% neutral
SentiBahasaRojak / BRCC [23]	**Five native Malaysian experts** able to read and write Bahasa Rojak	**Not stated**	**κ = 0.77, but only for the stock-review subset, and for a binary bullish/bearish judgement**	Whole of that subset; the augmented product and movie subsets carry no agreement figure	**No** class counts reported
MESocSentiment [3]	**Three software annotators; no human annotators**	**Not applicable; no human annotation stage**	**None; reliability is asserted ("three sentiment analysers were used to enhance the reliability") but never measured**	Manual check illustrated by example; no rate or sample size given	**Yes: 78.23% neutral, 16.33% positive, 5.44% negative**

Read down the columns, the audit yields four findings.

First, only one resource reports the full set, and it is the one that is not doing polarity. MY-EN-CS supplies all five elements, but its task is binary subjectivity, so the completeness of its reporting and the narrowness of its task go together. The three polarity resources each omit at least two elements, and no two omit the same combination, so their results are not comparable with one another even where they are individually sound.

Second, the two corpora that report no agreement statistic at all are not the ones expected. MESocSentiment is automatically labelled, so the absence is unsurprising. MyCovid-Senti is manually annotated by three people, works from a

fully specified four-question annotation guideline, and still reports no agreement figure. The assumption that manual annotation is self-evidently reliable is doing work here that the evidence does not support: manual annotation is more likely to be reliable, but reliability that is not measured has not been established.

Third, the reported coefficients do not measure what they appear to. The highest figure in the regional literature, $\kappa = 0.83$, is for a binary subjectivity decision, distinguishing opinionated from factual text. That is a different and easier task than three-class polarity, and binary tasks yield systematically higher agreement, so the figure cannot be read as evidence that polarity annotation on Malaysian code-mixed text achieves substantial agreement. The second figure, $\kappa = 0.77$, covers only the stock-review subset of SentiBahasaRojak, and it too is a binary judgement, bullish against bearish, on posts drawn from financial platforms rather than casual social media; the augmented product and movie subsets carry no figure at all. Neither coefficient the regional literature reports was therefore computed on the three-class polarity task these resources are used for. In both cases the number is sound; what is unsound is treating it as a reliability estimate for the resource as a whole. This is the sampling problem of Section 5.1 appearing in practice, and in a more severe form than that discussion anticipated, since here the estimate is drawn not from an unrepresentative sample of the target text but from a different register of it.

Fourth, and most directly relevant to Section 3.2, the class distributions permit a comparison the individual papers do not make. MyCovid-Senti, annotated by humans, is **23.6% neutral**. MESocSentiment, annotated by an automated ensemble, is **78.23% neutral**. Both are corpora of Malaysian tweets. The neutral share differs by a factor of more than three. The comparison is not clean, and the confound should be stated plainly: MyCovid-Senti is restricted to pandemic-related discourse, a domain in which strong feeling is more likely than in general conversation, and it was collected two years earlier. A difference of this magnitude and direction is nonetheless difficult to attribute to topic alone, and it is exactly what the mechanism described in Section 3.2 predicts. It is the closest thing the regional literature offers to a controlled observation that **the neutral skew is a property of the annotation method rather than of Malaysian discourse**.

The overall finding is therefore not that agreement in this field is low. It is that **agreement is frequently unmeasured, and where measured, measured under conditions that make the number difficult to interpret**: on a different task, on an unrepresentative subset, or on a distribution under which the coefficient behaves pathologically. The corpora whose labels are least examined are also, consistently, the ones produced by the methods most likely to introduce error.

One further deficiency is separable from the rest and worth stating on its own, because the literature routinely conflates the two. The absence discussed so far is a **corpus gap**: there is no large, balanced, casually authentic, human-validated resource to train on. Distinct from it is a **benchmark gap**: there is no shared, human-validated evaluation set for Malaysian code-mixed sentiment against which competing systems could be compared. The first concerns what exists to learn from, the second what exists to measure against. A field can close the first and remain unable to compare its own results, because every study continues to evaluate on its

own test split, drawn from its own corpus, labelled by its own procedure. Reported accuracies across these studies are consequently not commensurable, and the comparisons routinely drawn between them rest on an assumption of equivalence that no shared benchmark supports.

6. Critical Discussion

6.1. Synthesis of the Reviewed Literature

Sections 2 through 5 examined linguistic structure, annotation practice, model capability and evaluation largely in isolation, which is how the literature itself is organised. Table 4 consolidates the most representative works so that the patterns running across them become explicit. Its final column records what each work establishes about the reliability of its own labels; for the two corpus papers it summarises Table 3 rather than repeating it.

Table 4. Synthesis of Representative Approaches and Their Unfilled Gaps

Work	Domain and data	Method	Strength and unfilled gap	Evidence offered about its own label quality
Mohd Shamsuddin et al. [3]	Malay-English Twitter (MESocSentiment)	Semi-automated voting across a rule-based, a lexicon-based and a shallow transformer annotator	Scales annotation cheaply, **but** the voters share a blind spot, yielding 78.23% neutral, and contested cases are removed rather than resolved	None. Reliability asserted, never measured; manual checking illustrated by example only
Romadhona et al. [23]	Trilingual MS-EN-ZH (BRCC, SentiBahasaRojak)	Encoder with token-level language identification; augmentation for two of three subsets	Authentic Bahasa Rojak with switch-boundary awareness, **but** partly formal-register and augmentation-bound	$\kappa = 0.77$ for the stock-review subset only; no class distribution reported
Manukonda & Kodali [15]	Transliterated Malayalam, fake news detection	Transliteration-aware XLM-R, MLM over original and transliterated variants	Resolves phonetic and orthographic variation once it is represented in fine-tuning data, **but** coverage is bounded by that data	Not a corpus paper; reports task accuracy against existing labels

Work	Domain and data	Method	Strength and unfilled gap	Evidence offered about its own label quality
Negi et al. [8]	Fine-grained opinion, general domains	Generative LLMs as annotators, with an LLM adjudication stage	Annotates without task-specific training data, **but** performance bifurcates: reliable on spans, weak on relational structure	Reports its own bifurcation and concludes against full autonomy, positioning LLMs as assistants
Jadhav et al. [9]	Marathi, low-resource annotation	Frontier LLMs against fine-tuned BERT, head to head	Establishes that LLM annotators trail fine-tuned encoders by 10.2 and 14.1 accuracy points where resources are thin	Measures annotator output against existing gold labels; not a reliability study
Niimi [13]	English restaurant reviews (Yelp)	Quantised local LLM, majority vote across repeated attempts	Improves reproducibility, and multiple attempts with a medium model outperform one attempt with a large one, **but** aggregates a single model type	Addresses variability across trials of one annotator, not agreement between annotators

Three trends emerge from the table that are stated in none of the works individually.

Trend 1: flexibility trades against reliability. As annotation moves from static lexical matchers, through fine-tuned encoders, to zero-shot generative models, the dominant failure mode migrates rather than diminishing. Lexical tools fail by not parsing, and the output collapses to a default. Encoders fail by not covering, their competence ending where their fine-tuning data ends. Generative models fail by being unstable: output varies across identical queries [13] and relational structure is reproduced unfaithfully [8]. The property traded away at each step is the one that made the earlier failure easy to detect. A lexical tool that returns neutral for everything is transparently broken; a generative annotator producing fluent and inconsistent labels is not.

Trend 2: complementary capabilities are deployed in isolation. Fine-tuned encoders recognise regional patterns reliably but cannot interpret forms absent from their training data, while generative models supply that interpretive capacity and lose accuracy precisely where resources are thin. The capabilities are close to complementary, each strongest where the other is weakest. Yet every consensus mechanism in the reviewed literature aggregates voters drawn from a single competence class: Mohd Shamsuddin et al. [3] combine three annotators that all resolve sentiment from surface features, Niimi [13] combines repeated samples of one model, and Negi et al. [8] adjudicate between outputs of the same kind of annotator. The premise that pooling annotators cancels their errors has therefore been tested only under the condition least likely to satisfy it.

Trend 3: reliability is under-evidenced relative to construction. Across the corpus papers, the effort invested in producing labels substantially exceeds the effort invested in showing them to be sound: elaborate pipelines, and in one case a carefully documented guideline, accompanied by either no agreement statistic or one measuring a narrower task than the resource is used for. This is a property of the field's reporting conventions rather than of any individual study, which is why it is visible only in aggregate.

From these trends, three gaps follow.

Gap 1: the fidelity-scalability trade-off remains unresolved. Manual annotation yields defensible labels at volumes and topical coverage insufficient for a general-purpose resource, while single-layer automation reaches scale by defaulting to neutral on the material that most needs judgement. No reviewed strategy secures both (Section 3.1), and no existing corpus is simultaneously large-scale, polarity-balanced, casually authentic and human-validated (Section 5.2).

Gap 2: cross-competence consensus is untested. The literature establishes what encoders and generative models each do well, and nothing about what happens when annotators of genuinely different competence are required to agree. What is missing is a measurement rather than a specification for a system: whether annotators of different types fail on the same inputs determines whether aggregating them would help at all. Section 3.2 identified the independence premise on which the affirmative answer depends and noted that it is nowhere tested on this material.

Gap 3: reliability is neither consistently reported nor, under the prevailing class skew, reliably measurable. Two components are usually conflated. The first is a reporting failure and is straightforwardly remediable: agreement statistics, annotator qualifications, guidelines, sampling schemes and class distributions are reported unevenly and incompletely. The second is a measurement problem that better reporting cannot fix, because on a corpus as skewed as MESocSentiment the standard coefficient does not behave interpretably however scrupulously it is reported. Section 6.2 shows that this gap is not independent of the other two.

6.2. How the Challenges Compound

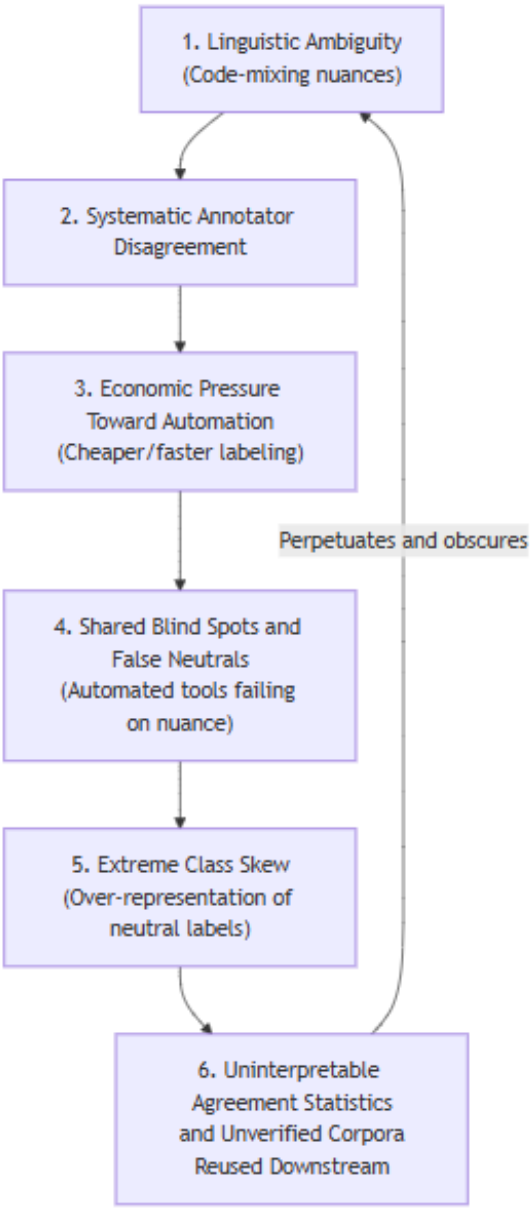


Figure 1. The Compounding Chain

The three gaps are conventionally treated as separate problems with separate remedies: better corpora for the first, better annotator ensembles for the second, better reporting for the third. They are not separate. Each degrades the conditions on which the next depends, and the sequence closes on itself, which is why interventions directed at any single point have not resolved the problem. What follows concerns only the relations between the findings of Sections 2 to 5, which no reviewed study is positioned to observe because each addresses one link.

Link 1: from linguistic structure to systematic disagreement. Switching in this variety is intrasentential, and affect is frequently carried by particles absent from both standard lexicons (Section 2). The consequence is not simply that agreement is harder to achieve but that **disagreement takes a particular form**. Where an annotator's reading of lah as softener rather than sarcasm marker is stable across occurrences, the disagreement recurs on the same construction in the same direction. Random disagreement is reduced by recruiting more annotators, since independent errors average out; systematic disagreement is not, because additional annotators bring their own stable readings rather than noise around a shared one. The field's default response to unreliable annotation is therefore structurally unsuited to the kind of unreliability this text produces.

Link 2: from disagreement to automation. The reviewed corpus papers consistently cite cost and annotator availability as the reason for automating. Mohd Shamsuddin et al. [3] state that manual classification "would be laborious and costly" and that automation "eliminated the need to find speakers fluent in Malay and English"; Negi et al. [8] open on the "considerable human effort and substantial cost" of manual opinion annotation; Niimi [13] cites the expense of crowdsourced labelling directly. What the papers do not state, and what this review reads as causal, is the connection back to Link 1: the annotator who must be recruited is not merely a human but a fluent bilingual one, and that requirement follows from the linguistic properties of the text rather than from any general fact about annotation. Automation on this material is therefore downstream of the linguistic difficulty rather than an independent methodological preference.

Link 3: from automation to skew. Annotators selected under cost pressure are selected for availability, and the available tools are those built for well-resourced English. Because those tools resolve sentiment from surface lexical matching, they share a blind spot rather than holding independent ones, so pooling them ratifies the shared error instead of cancelling it (Section 3.2). The false neutrals accumulate rather than being corrected, and the corpus acquires an extreme class distribution that no participant in the process intended or, in the absence of a reliability measurement, necessarily noticed.

Link 4: the closing loop. Links 1 to 3 describe a corpus acquiring a defect; this link concerns what happens when someone attempts to detect it, and it is the point at which the review's two halves meet. The skew produced at Link 3 is the specific condition under which the field's standard reliability instrument stops returning interpretable values. On a heavily dominated class distribution, expected chance agreement rises toward observed agreement, so kappa collapses while raw percent agreement inflates (Section 5.1). A researcher who suspects the labels are unsound and computes kappa obtains a low value, consistent with poor annotation and equally consistent with competent annotation of a skewed corpus; computing raw agreement yields a high value carrying the same ambiguity in the opposite direction. **The defect and the inability to detect the defect have a common cause.** They are not two problems that happen to co-occur but one condition described twice, and that is what separates an unresolved problem from a merely difficult one. In a merely difficult problem the quality check works and reports bad news; here the

quality check is least informative precisely where quality is lowest, so a defective corpus and a sound one are, by the field's standard measure, hard to tell apart.

Link 5: the perverse incentive. Section 3.3 described the removal of posts on which the annotators could not reach a majority: 512 items, 2.53% of those labelled, withdrawn on the stated grounds of retaining only "absolute categories". Those items are not a random sample of the corpus; they are disproportionately the contested, heavily code-switched cases that Link 1 identifies as the source of the difficulty. Removing them raises measured agreement while lowering representativeness, and in this instance it also made the distribution worse: neutral rose from 76.23% of the labelled set to 78.23% of what remained. The pattern is not confined to automated pipelines: Kasmuri and Basiron [5] annotated 4,191 sentences and carried 2,316 into their corpus, discarding the remainder, among them the sentences their two annotators labelled differently. A procedure whose reported reliability improves as it deletes the data that would have tested it deserves to be named as such, and nothing in current reporting practice would expose it, since a corpus paper reports the corpus it kept rather than the corpus it discarded.

Link 6: propagation and the return edge. Corpora whose reliability was never established are released, trained on, evaluated against and cited, so every downstream comparison inherits an error term that no one has quantified, and differences between reported system accuracies cannot be separated from differences in the labels they were measured against. The edge that closes the cycle is the more consequential one: the annotators used to build the next corpus are trained or fine-tuned on the last, so the blind spots of one generation of labels are transmitted to the instruments that will produce the next. The chain does not merely persist; **it reproduces itself, and each cycle makes the shared blind spot more entrenched and less visible**.

Read as a whole, the chain explains a pattern that the individual studies cannot. Each reviewed work intervenes at one link, competently: better normalisation at Link 1, cheaper annotation at Link 2, more voters at Link 3, more careful reporting at Link 4. **No reviewed work addresses the chain as a chain**, and an intervention at one link is assessed by an instrument compromised at another, so the assessment cannot establish whether the intervention worked. Establishing that this structure exists is the contribution this review claims.

6.3. Directions for Future Research

The chain in Section 6.2 identifies where intervention would matter, not which intervention would succeed. The directions below are accordingly framed as work the field should undertake rather than as a programme any particular study should adopt, and each is traced to a gap or link established earlier, since a direction that cannot be traced to a finding is a preference rather than a conclusion.

Direction 1: reporting standards for code-mixed sentiment corpora. *(From Gap 3.)* The most immediately actionable deficiency is also the least technically demanding to correct. A minimum reporting set for any newly released corpus would comprise the number of annotators and their stated qualification, the availability of the annotation guideline, the sampling scheme behind any agreement

estimate, per-class rather than aggregate agreement, and the full class distribution. Table 3 shows how unevenly these are currently supplied: only the binary-subjectivity resource reports all five, the three polarity resources each omit at least two, and the omissions differ between them, so results cannot be compared even where they are individually sound. Per-class agreement deserves particular emphasis, since an aggregate figure on a skewed corpus is dominated by the majority class and can conceal near-total disagreement on the minority polarities that matter most.

Direction 2: agreement measurement under extreme class skew. *(From Link 4.)* Prevalence-robust coefficients are established in principle [22] but have not been evaluated on code-mixed sentiment data, so it is not known which behaves best under the distributions these corpora actually exhibit. The study is tractable and self-contained: an existing annotated corpus, a set of candidate coefficients, and a systematic comparison of their behaviour as skew increases, rather than new data collection. It is also a precondition for the others, since until the field can measure reliability under its own prevailing conditions, no corpus built hereafter can be shown to be an improvement on the ones criticised here.

Direction 3: guideline design for affect-bearing function words. *(From Link 1.)* Because the disagreement generated by discourse particles is systematic rather than random, the appropriate remedy is a shared annotation convention rather than additional annotators, and the field has none for lah, kan and their equivalents. A general model exists: Mohammad [24] proposes structured schemes that decompose the sentiment judgement into separate questions about the speaker's emotional state, the target of the opinion and the attitude toward it, and that treat sarcasm, ridicule and mockery as a distinct response option rather than folding them into polarity. That provision addresses the specific ambiguity identified in Section 2, where the same particle admits a softening and a sarcastic reading. Kong et al. [6] adapt precisely this questionnaire, citing Mohammad directly, into a four-question semantic-role guideline covering the speaker's emotional state, the primary target of opinion, attribution of quoted speech and majority public sentiment. The design pattern is therefore not merely available but already implemented in the region. What is absent is not the design pattern but its adaptation to Malaysian discourse particles and its adoption across projects.

Direction 4: treating disagreement as data rather than as error. *(From Link 5.)* Current practice resolves annotator disagreement by majority rule or removes it from the corpus, in both cases discarding the information that the case was contested. Retaining and publishing the full label distribution for each item, rather than only the resolved label, would preserve the ambiguous cases that present pipelines delete and would allow ambiguity to be studied rather than eliminated. It would also make Link 5 observable: a corpus that reports what it discarded, and on what grounds, allows a reader to judge whether its reported reliability was earned or obtained by attrition.

Direction 5: testing the independence assumption. *(From Gap 2.)* Every consensus mechanism in this literature depends on the premise that annotators of different construction fail independently, since aggregation cancels error only to the extent that this holds, and on code-mixed input the premise is asserted throughout

and measured nowhere. The study the field lacks is a direct one: label a common sample with annotators of deliberately different competence, encoder-based, generative and human, and measure the correlation of their errors rather than their individual accuracies. Low correlation would justify the aggregation strategies already in use; high correlation would indicate that consensus on this material adds confidence without adding evidence. Either result changes what the field should build next, which is why this should precede rather than follow further work premised on the assumption.

Direction 6: targeted rather than uniform human validation. *(From Gap 1.)* The fidelity-scalability trade-off is treated in the literature as though human effort must be applied uniformly or not at all. Whether it can instead be routed to the cases where automated labels are least reliable, and at what cost per corrected label, is unexamined. The question is empirical and has a measurable answer: it requires a criterion for identifying low-confidence labels, a measurement of how often correction is warranted at each confidence level, and an accounting of the annotator time consumed. A negative result would be informative, since it would establish that the trade-off is genuinely binding rather than merely unaddressed.

Direction 7: a shared evaluation benchmark. *(From Section 5.2.)* The benchmark gap is distinct from the corpus gap and is not closed by building better corpora. Comparability across studies requires a common, human-validated test set for Malaysian code-mixed sentiment, held separate from any single project's training data and stratified so that heavily code-switched items are represented in known proportion rather than in whatever proportion a random sample yields. Its construction is a field-level undertaking rather than a by-product of individual projects, which is the likeliest reason it does not yet exist.

These are directions rather than results. The claim of this review is not that the interventions above will succeed, but that the structure described in Section 6.2 identifies where the field's effort is currently being absorbed without accumulating, and that the second direction in particular is a precondition for knowing whether any of the others have worked.

7. Conclusion

This review examined a single question: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data remains unresolved. The answer offered is that the field has been diagnosing the wrong constraint. Underperformance on this material is routinely attributed to model capacity; the evidence assembled here locates the binding constraint upstream, in the quality of the labels and the credibility of the evidence offered that they are sound.

The central finding is contained in one relation. **The class skew that corrupts the labels is the same condition under which the standard agreement statistic ceases to be interpretable, so the defect disables the instrument that would have detected it.** This is what distinguishes an unresolved problem from a merely difficult one. A difficult problem is one where the quality check works and returns bad news; here the quality check is least informative precisely where quality is lowest, which is why competent interventions at individual points in the chain have

accumulated without the field being able to demonstrate that anything has improved.

Several questions follow and remain open. Whether adjudication of annotator disagreement is affordable at corpus scale is unknown, its cost having been reported only for small evaluations. Whether a shared annotation convention for Malaysian discourse particles can reduce the systematic component of disagreement has not been attempted. Whether any agreement coefficient behaves interpretably under the class distributions these corpora actually exhibit has not been tested. And whether annotators of different construction fail independently on code-mixed input, the premise on which every consensus method in this literature rests, is asserted throughout and measured nowhere.

The last of these determines whether the aggregation strategies already in use are sound; the third determines whether the field could tell. Until it is established which coefficient behaves interpretably under these conditions, a corpus that has genuinely improved and one that has merely been measured more favourably will remain difficult to tell apart, and that is the question on which the others depend.

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